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The relationship between problematic social media use and psychological factors among university students with evaluation of ADHD, social anxiety and self-esteem

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Abstract

Problematic social media use (PSMU) has been increasingly associated with various psychological vulnerabilities; however, findings remain mixed, particularly among university students and in non-Western contexts. This study aimed to examine the associations between PSMU and psychological factors including attention-deficit/hyperactivity disorder (ADHD) symptoms, social anxiety, and self-esteem among university students, as well as to explore the role of selected sociodemographic and usage-related characteristics. This cross-sectional study included 179 university students aged 18 to 25 years. Participants completed a sociodemographic and clinical information form, the Social Media Addiction Scale–Adult Form, Adult Attention-Deficit/Hyperactivity Disorder Self-Report Scale, Liebowitz Social Anxiety Scale, and Rosenberg Self-Esteem Scale–Short Form. Statistical analyses were conducted using SPSS v22.0. Spearman correlation analyses and group comparisons were performed, with statistical significance set at $P < .05$. Mean daily screen time was 352.29 ± 156.41 minutes, significantly higher than the global average. Social media addiction scale scores positively correlated with ADHD symptoms ($r = 0.412$, $P < .001$), social anxiety ($r = 0.395$, $P < .001$), and low self-esteem ($r = 0.221$, $P = .03$). Participants who found it easier to communicate on social media exhibited higher social anxiety scores. The use of social media with hidden identities or fake accounts was associated with increased screen time, social media addiction severity, and social anxiety. The findings indicate that problematic social media use among university students is closely associated with psychological vulnerabilities, particularly ADHD symptoms and social anxiety, whereas the role of self-esteem appears more nuanced. These results support the conceptualization of PSMU as a multidimensional phenomenon influenced by individual psychological traits and usage-related behaviors rather than by time spent online alone. By providing data from a non-Western university student population, this study contributes to the cultural generalizability of existing literature and highlights the importance of considering multiple psychological domains when addressing problematic social media use.

Abbreviations: ADHD = attention-deficit/hyperactivity disorder, ASRS = Adult Attention-Deficit/Hyperactivity Disorder Self-Report Scale, LSAS = Liebowitz Social Anxiety Scale, PSMU = problematic social media use, RSES = Rosenberg Self-Esteem Scale-Short Form, SMAS-AF = Social Media Addiction Scale-Adult Form.

Keywords: ADHD, PSMU, self-esteem, social anxiety, social media

1. Introduction

The exponential growth of digital technologies has led to a marked increase in global internet usage, with over 62.6% of the population actively engaging with social media platforms as of May 2024.[1,2] Despite its ubiquity, the empirical literature on problematic social media use (PSMU) remains underdeveloped relative to broader research on internet and smartphone addiction. Internet addiction presents in various subtypes, including social media addiction, gaming disorder, cybersex addiction, and compulsive online shopping.[3] While the detrimental effects – such as social withdrawal, financial distress, familial conflict, and deviant behavior in minors – are well documented, the nosological status of internet-related disorders remains contentious. Competing paradigms situate these behaviors within the frameworks of substance use disorders, obsessive-compulsive spectrum disorders, or impulse control disorders, reflecting significant diagnostic ambiguity.[4,5] Consequently, terminological inconsistency persists, with overlapping labels such as "problematic internet use," "internet addiction and "internetomania."[6,7]

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The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Andreassen and Pallesen define PSMU as excessive preoccupation with and compulsive use of social networking platforms, impairing psychosocial functioning.[8] Other terms used to describe this condition include "Social Media Use Disorder and "Networks Use Disorder."[9] While usage alone may be neutral or modestly linked with mental health outcomes, PSMU – defined by compulsive, addiction-like behaviors (e.g., preoccupation, withdrawal, tolerance, loss of control) – has emerged as a more clinically relevant construct.[10]

Excessive social media use, similar to excessive internet or smartphone use, has been linked to various issues, including sleep quality,[11,12] symptoms of anxiety,[13,14] depression,[13,15,16] reduced psychological well-being,[17] diminished self-esteem[13] and psychological distress.[18,19] Nevertheless, alongside evidence of adverse psychological effects, some studies highlight potential benefits, including enhanced quality of life, improved social support, greater well-being and decreased stress.[20,21] Problematic social media use refers to patterns that mirror behavioral addiction, characterized by compulsive engagement and an inability to reduce usage despite negative consequences.[22] Like heavy usage, problematic use has been associated with mental health problems, including increased symptoms of depression and anxiety, lower well-being, and reduced self-esteem.[23-26] A focused meta-analysis of 53 studies revealed a robust positive correlation between social anxiety and PSMU.[10] Studies have demonstrated a positive correlation between problematic social networking site use and attention deficit hyperactivity disorder (ADHD).[11,27] Further studies suggest links to major depressive disorder, obsessive-compulsive disorder, eating disorders, and substance use.[28] A study on personality traits found a positive correlation between impulsivity and social media addiction, while self-esteem served as a protective factor.[29] Notably, social media can be beneficial for individuals with low self-esteem and self-confidence by providing support and facilitating communication within their social circles.[30]

Recent research has increasingly examined the relationship between ADHD and social media use, highlighting both frequent and problematic patterns of engagement.[31,32] Studies across various digital domains, including television and internet use, suggest a temporal association between early media exposure and later ADHD symptoms, though these studies often lack clinical ADHD samples.[33] Meta-analyses reveal modest links between ADHD traits and digital engagement, especially internet addiction, but do not support strong causal inferences. Empirical findings further differentiate between intensity of use – frequent use not necessarily indicating dysfunction – and problematic use, which more consistently correlates with ADHD symptoms.[34-36] Adolescents with ADHD tend to use social media more intensely, hold more accounts (also more fake accounts) and exhibit higher distractibility and impulsivity online.[37-39] The studies found an association between PSMU and ADHD characteristics. In a large survey, adolescents reported an association of addictive social media use with hyperactivity.[40] A study showed that, users with ADHD have significantly different posting behaviors, such that they posted more tweets and used more swear words.[41] The studies mostly focused on the Facebook and Twitter as social media channels.[42]

Social anxiety is widely recognized as a significant risk factor PSMU.[43,44] Individuals with high social anxiety or social deficits may overuse social media to compensate for offline social difficulties.[45] Social media provides a comfortable virtual environment, potentially replacing real-world social interactions. PSMU acts as a coping mechanism to fulfill unmet social needs among those with elevated social anxiety.[46-48] Conversely, socially skilled and extraverted individuals also engage intensively with social media, thereby increasing PSMU risk, indicating that the link between social anxiety and PSMU is not straightforward.[49-51] Empirical studies report positive correlations between social anxiety and PSMU, but effect sizes vary substantially, with some studies finding no significant association.[10] Gender differences remain inconclusive: some research indicates males are more prone to PSMU. Some studies have even suggested that the relationship between social anxiety and PSMU may be stronger in females than in males, whereas other studies suggest minimal or even stronger effects in females.[52-54] Age as a moderator is also unclear, and cultural factors also play an important role.[49,55]

The studies suggested that low self-esteem and PSMU are consistently correlated, but the causal direction of their relationship remains unclear.[56] Two dominant perspectives exist. The first one suggests that low self-esteem contributes to PSMU. Individuals engage with media to fulfill psychological and social needs.[57] Personal traits – such as self-esteem – influence susceptibility to internet-related behavioral addictions. Individuals with low self-esteem often experience social difficulties and a weak sense of self, making them more likely to seek affirmation through social media. This behavioral pattern may lead to PSMU as users increasingly rely on digital interactions for social validation and identity reinforcement.[58] Conversely, the second perspective contends that PSMU may negatively impact self-esteem.[30] Frequent social media use encourages individuals to experiment with multiple online personas, weakening their sense of a coherent identity.[59,60] Furthermore, upward social comparisons, common on networking platforms, can intensify feelings of inferiority, contributing to lower self-esteem.[61]

Most research in this area relies on cross-sectional data, limiting conclusions regarding directionality rather than causality, and has predominantly focused on adolescent samples.[56,62] Although university students are often classified as late adolescents, they represent a distinct developmental and contextual group characterized by increased autonomy, reduced parental monitoring, and greater exposure to academic and social stressors. Research examining the associations between problematic social media use and psychological factors specifically among university students, particularly in non-Western countries, therefore remains relatively limited. This study aims to examine the associations between problematic social media use and psychological factors including ADHD symptoms, social anxiety, and self-esteem among university students. In addition, the study explores the role of usage-related characteristics (screen time, communication preferences, and anonymous account use) in relation to problematic social media use.

2. Materials and methods

2.1. Participants

This study was conducted at the Department of Psychiatry, TOBB University of Economics and Technology, and received ethical approval from the local ethics committee (Ethics Committee No. KAEK-118/134). The study included 179 university students aged 18 to 25 who met the inclusion criteria and provided informed consent. A convenience sampling method was used. Participants were recruited through announcements on the university's official faculty communication platform. After receiving ethics approval, a brief explanation of the study, the informed consent form, and a secure survey link were sent to eligible students. Inclusion criteria were being within the specified age range and willing to participate. Exclusion criteria included reluctance to participate, incomplete participation, or expectation of financial benefit. Data were recorded anonymously.

2.2. Procedure

This study was designed as a cross-sectional one. The participants were assessed with a sociodemographic and clinical form, the Social Media Addiction Scale- Adult Form (SMAS-AF), Adult Attention-Deficit/Hyperactivity Disorder Self-Report

Scale (ASRS), Liebowitz Social Anxiety Scale for Adults (LSAS), Rosenberg Self-Esteem Scale- Short Form (RSES). All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all individual participants included in the study.

2.3. Instruments

Participants were assessed using the following instruments:

The Sociodemographic and Clinical Information Form: Sociodemographic data (age, gender, social environment, income status), screen time, anxiety level when unable to access social media, and social media use patterns were collected. Anxiety level when unable to access social media was assessed via self-report using a visual analog scale ranging from 0 (no anxiety) to 10 (extreme anxiety).

The Social Media Addiction Scale-Adult Form (SMAS-AF): The validity and reliability of which was made by Şahin and Yağcı,[63] is a scale consisting of 2 sub-dimensions and a total of 20 items. The score for each question is rated from 1 to 5, "1-Not suitable for me at all," "2-Not suitable for me," "3- Undecided," "4-Suitable for me," "5-Very suitable for me." Negative expressions were reverse coded. The virtual tolerance sub-dimension consists of items 1 to 11, and the virtual communication subdimension consists of items 12 to 20. The lowest score that can be obtained from the scale is 20, and the highest score is 100. The Cronbach α internal consistency coefficient for the overall scale was reported as 0.94.

Adult Attention-Deficit/Hyperactivity Disorder Self-Report Scale (ASRS): This scale was developed by the World Health Organization to detect attention deficit and hyperactivity disorder in adults. The validity and reliability study was conducted by Doğan et al., and the Cronbach α was 0.8.[64] It is an 18-item self-report scale consisting of 9 items each, with 2 subscales: attention deficit (A) and hyperactivity/impulsivity (B). The score of each question is rated from 0 to 4 (never = 0, rarely1, sometimes = 2, often = 3 points, and very often = 4 points). A score of 24 or above on either of the 2 subscales is indicative of attention-deficit/hyperactivity disorder. The use of the ASRS in this study reflects its suitability as a screening tool for assessing ADHD symptoms in non-clinical populations. However, it should be noted that the scale does not provide a clinical diagnosis, and the results should therefore be interpreted as reflecting symptom severity rather than confirmed ADHD.

The Liebowitz Social Anxiety Scale for Adults (LSAS): The scale, developed in 1987 by Michael Liebowitz, consists of 24 items and 2 subscales. The 24 items are first rated on a Likert scale from 0 to 3 on fear felt during the situations, and then the same items are rated regarding avoidance of the situation (Fear: none = 0, mild = 1, moderate = 2, severe = 3 points/ Avoidance: never = 0, occasionally = 1, often = 2, usually = 3 points). The total scores for the Fear and Avoidance sections provide an overall score with a maximum of 144 points. The validity and reliability study was conducted by Dilbaz and Güz.[65] The Cronbach α was 0.98.

Rosenberg Self-Esteem Scale-Short Form (RSES): The scale, which is considered a reference in self-esteem measurement, was developed by Morris Rosenberg in 1963. The reliability and validity studies of the scale in Turkey were conducted by Çuhadaroğlu (1986), and the Cronbach α was 0.71.[66] The scale consists of twelve subdomains, and the first ten items measure self-esteem. Items 1, 2, 4, 6, 7 are positively coded, while items 3, 5, 8, 9, and 10 are negatively coded. A low score in the scale indicates high self-esteem; a high score indicates low self-esteem.

2.4. Statistical analysis

Statistical analysis was performed using SPSS (Statistical Package for Social Sciences) version 22.0 for Windows. Mean ± standard deviation and median (minimum–maximum) were used as descriptive statistics for quantitative variables, and the number of people (percentage) was used for qualitative variables. Normality of distribution of the numerical variables was examined by the Kolmogorov–Smirnov test, and the homogeneity of the variances was examined by the Levene test. When data were not normally distributed, nonparametric tests were applied. Spearman correlation analysis was used to examine relationships between numerical variables. Participants were categorized into 3 groups based on how they compared social media with face-to-face communication (more difficult on social media, no difference, easier on social media) and into 2 groups based on whether they used social media with a hidden identity or fake account. The Mann–Whitney *U* test was used for 2-group comparisons, and the Kruskal–Wallis test was used for comparisons of more than 2 independent groups. Categorical variables were compared using the chi-square test. The significance level was set at $P < .05$.

3. Results

3.1. Demographic and clinical characteristics

One hundred seventy-nine university students between the ages of 18 to 25 participated in this study. Demographic and clinical characteristics of the study population are presented in Table 1. The sample consisted of 130 women (72.6%) and 49 men (27.4%). The mean age of the sample was 20.58 ± 1.75 (median = 20) years. It was observed that 46.4% of the participants had more income than their expenses, and 96.1% of them grew up in urban areas. The mean daily screen time was 352.29 ± 156.41 minutes (median = 330), with Instagram being the most used social media platform. Mean scores of the clinical scales (SMAS-AF = 55.40 ± 11.97, ASRS = 34.41 ± 11.80, LSAS = 97.25 ± 25.83, RSES = 12.95 ± 7.38) are presented in Table 1 with subscale scores. The frequencies related to subjective perceptions and thoughts about social media are shown in Table 2.

3.2. Relationship of clinical scale scores with other clinical variables

The relationship between social media use pattern/ addiction severity and the clinical variables of the participants were studied (Table 3). Screen time had a positive correlation with anxiety level when unable to access social media ($r = 0.317$, $P < .001$), total score of SMAS-AF ($r = 0.32$, $P < .001$), SMAS-AF-virtual tolerance scores ($r = 0.391$, $P < .001$), SMAS-AF- virtual communication scores ($r = 0.179$, $P = .016$), total score of ASRS ($r = 0.158$, $P = .034$), ASRS-A scores ($r = 0.196$, $P = .009$), total score of LSAS ($r = 0.294$, $P < .001$), LSAS-Fear scores ($r = 0.277$, $P < .001$), LSAS-Avoidance scores ($r = 0.274$, $P < .001$); there was no correlation between screen time and ASRS-B scores ($r = 0.109$, $P = .147$), total score of RSES ($r = 0.106$, $P = .157$).

Total scores of SMAS-AF had a positive correlation with total scores of ASRS ($r = 0.412$, $P < .001$), ASRS-A scores ($r = 0.44$, $P < .001$), ASRS-B scores ($r = 0.254$, $P = .001$), total score of LSAS ($r = 0.395$, $P < .001$), LSAS-Fear scores ($r = 0.382$, $P < .001$), LSAS-Avoidance scores ($r = 0.366$, $P < .001$), and total scores of RSES ($r = 0.221$, $P = .03$). Also, SMAS-AF-virtual tolerance and SMAS-AF-virtual communication subscale scores had same positive correlations except that there was no correlation between SMAS-AF-virtual communication subscale score and total score of RSES.

The anxiety level when unable to access social media had a positive correlation with total scores of SMAS-AF ($r = 0.468$,

$P < .001$), SMAS-AF-virtual tolerance scores ($r = 0.492$, $P < .001$), SMAS-AF-virtual communication scores ($r = 0.338$, $P < .001$), total score of ASRS ($r = 0.233$, $P = .002$), ASRS-A scores ($r = 0.213$, $P = .004$), ASRS-B scores ($r = 0.216$, $P = .004$), total score of LSAS ($r = 0.181$, $P = .015$), LSAS-Fear scores ($r = 0.17$, $P = .023$), LSAS-Avoidance scores ($r = 0.166$, $P = .026$), and total scores of RSES ($r = 0.065$, $P = .384$).

When we looked at it according to gender, it was seen that there was a significant difference in the total scores of SMAS-AF ($P = .02$), SMAS-AF-virtual tolerance scores ($P = .01$), SMAS-AF-virtual communication scores ($P = .016$), total score of LSAS ($P = .006$), LSAS-Fear scores ($P = .001$), LSAS-Avoidance scores ($P = .022$).

Table 1

Sociodemographic and clinical scale scores of participants.

	n (%)	
Gender		
Female/male	130 (72.6)/49 (27.4)	
Social environment		
Urban/rural	172 (96.1)/7 (3.9)	
Income status		
Less than expenses/equal to expenses/more than expenses	21 (11.7)/67 (37.4)/83 (46.4)	
History of mental distress (lifetime)		
Yes/no	76 (42.5)/103 (57.5)	
Current psychotropic medication use		
Yes/no	20 (11.2)/159 (88.8)	
	Mean ± SD	**Median***
Age	20.58 ± 1.75	20
Daily screen time (minutes)	352.29 ± 156.41	330
Instagram	91.84 ± 76.32	77
WhatsApp	62.04 ± 55.49	50
YouTube	73.28 ± 79.16	50
Twitter (X)	17.80 ± 33.96	0
Facebook	2.85 ± 27.54	0
Clinical scales		
SMAS-AF Total	55.40 ± 11.97	
Virtual tolerance	34.28 ± 7.04	
Virtual communication	21.12 ± 6.24	21
ASRS Total	34.41 ± 11.80	
A-attention deficit	17.54 ± 6.49	
B-hyperactivity/impulsivity	16.87 ± 6.61	16
LSAS Total	97.25 ± 25.83	96
Fear	50.53 ± 13.47	49
Avoidance	46.72 ± 13.61	45
RSES	12.95 ± 7.38	13

ASRS = Adult ADHD Self-Report Scale, LSAS = Liebowitz Social Anxiety Scale: For Adults, RSES = Rosenberg Self-Esteem Scale-Short Form, SD = standard deviation, SMAS-AF: Social Media Addiction Scale-Adult Forum.

*Median was given for data with no normal distribution.

3.3. Comparison of clinical variables in groups separated by social media usage patterns

The participants were evaluated according to their comparison between communication on social media and face-to-face communication and were divided into 3 groups (more difficult on social media than face-to-face/ no difference/ easier on social media than face-to-face). Participants in groups were compared in terms of gender, age, screen time, the anxiety level score when not able to access social media, total scores of SMAS-AF, SMAS-AF-virtual tolerance scores, SMAS-AF-virtual communication scores, total score of ASRS, ASRS-A scores, ASRS-B scores, total score of LSAS, LSAS-Fear scores, LSAS-Avoidance scores, total score of RSES. There was a statistically significant difference in terms of the anxiety level when unable to access social media ($P = .024$), total scores of ASRS ($P = .035$), ASRS-A scores ($P = .024$), total score of LSAS ($P = .004$), LSAS-Fear scores ($P = .014$), LSAS-Avoidance scores ($P = .004$). There were no significant differences in gender, age, screen time, total scores of SMAS-AF, SMAS-AF-virtual tolerance scores, SMAS-AF-virtual communication scores, ASRS-B scores, total score of RSES between the groups (Table 4).

The participants were evaluated for the presence of social media use by hiding their identity or using a fake account and were divided into 2 groups. Participants in both groups were compared in terms of gender, age, screen time, anxiety level score when unable to access social media, total score of SMAS-AF, SMAS-AF virtual tolerance scores, SMAS-AF virtual communication scores, total score of ASRS, ASRS-A scores, ASRS-B scores, total score of LSAS, LSAS-Fear scores, LSAS-Avoidance scores, total score of RSES. A statistically significant difference was observed in gender ($P = .035$), screen time ($P = .039$), total score of SMAS-AF ($P < .001$), SMAS-AF virtual tolerance scores ($P = .002$), SMAS-AF virtual communication scores ($P = .003$), ASRS-A scores ($P = .027$), total score of LSAS ($P = .017$), LSAS-Fear scores ($P = .042$), LSAS-Avoidance scores ($P = .017$). There

Table 2

Participants' thoughts and perceptions on social media.

	Never	**Rarely**	**Sometimes**	**Often**	**Always**
Frequency of checking social media	1.7%	4.5%	25.1%	52%	16.8%
Need to check social media first thing when you wake up in the morning	5%	17.9%	28.5%	31.3%	17.3%
Frequency of argument on social media	51.4%	30.7%	12.8%	2.8%	2.2%
	Not having a fake account	**Harder**	**No difference**	**Easier**	
Comparison of communication with fake account with real one	57.9%	0.6%	27%	%14.6	
The effect of social media	**Negative effect**	**No effect**	**Positive effect**		
Your life	59.1%	15.9%	25%		
Academic success	64%	27.5%	8.4%		
Family relationships	33.5%	48%	18.4%		
Friendship	16.8%	21.8%	61.5%		
Romantic relationships	26.8%	41.3%	31.8%		
Comparison of communication on social media with face-to-face	**More difficult**	**No difference**	**Easier**		
Need to cut down the social media	41.9% *Yes* 61.8%	19% *No* 38.2%	39.1%		

Table 3

The relationship between clinical scale scores and clinical variables.

	Age	Screen time	Anxiety point unable to access (0–10)	SMAS-AF Total	SMAS-AF VT	SMAS-AF VC	ASRS Total	ASRS-A	ASRS-B	LSAS Total	LSAS Fear	LSAS Avoidance	RSES	Gender* Female/male (mean ± SD)
Age														*20.55 ± 1.72/20.59 ± 1.85* *P = .982*
Screen time	*r = 0.001* *P = .989*													*359.1 ± 153.82/336.67 ± 160.99* *P = .276*
Anxiety point unable to access (0–10)	*r = –0.016* *P = .827*	***r = 0.317*** ***P < .001***												*5.81 ± 2.6/4.76 ± 2.67* *P = .028*
SMAS-AF Total	*r = –0.022* *P = .769*	***r = 0.32*** ***P < .001***	***r = 0.468*** ***P < .001***											*57 ± 11.39/50.55 ± 12.49* ***P = .02***
SMAS-AF VT	*r = 0.047* *P = .531*	***r = 0.391*** ***P < .001***	***r = 0.492*** ***P < .001***	***r = 0.913*** ***P < .001***										*35.34 ± 6.65/31.27 ± 7.26* ***P = .01***
SMAS-AF VC	*r = –0.04* *P = .599*	***r = 0.179*** ***P = .016***	***r = 0.338*** ***P < .001***	***r = 0.866*** ***P < .001***	***r = 0.583*** ***P < .001***									*21.66 ± 6.27/19.29 ± 6.08* ***P = .016***
ASRS Total	*r = 0.03* *P = .692*	***r = 0.158*** ***P = .034***	***r = 0.233*** ***P = .002***	***r = 0.412*** ***P < .001***	***r = 0.417*** ***P < .001***	***r = 0.223*** ***P = .003***								*34.98 ± 11.67/32.57 ± 12.05* *P = .091*
ASRS- A	*r = 0.07* *P = .36*	***r = 0.196*** ***P = .009***	***r = 0.213*** ***P = .004***	***r = 0.44*** ***P < .001***	***r = 0.461*** ***P < .001***	***r = 0.243*** ***P = .001***	***r = 0.898*** ***P < .001***							*17.83 ± 6.46/16.47 ± 6.6* *P = .087*
ASRS- B	*r = –0.017* *P = .821*	*r = 0.109* *P = .147*	***r = 0.216*** ***P = .004***	***r = 0.254*** ***P = .001***	***r = 0.292*** ***P < .001***	***r = 0.169*** ***P = .021***	***r = 0.903*** ***P < .001***	***r = 0.604*** ***P < .001***						*17.15 ± 6.69/16.1 ± 6.37* *P = .269*
LSAS Total	*r = –0.075* *P = .319*	***r = 0.294*** ***P < .001***	***r = 0.181*** ***P = .015***	***r = 0.395*** ***P < .001***	***r = 0.401*** ***P < .001***	***r = 0.284*** ***P < .001***	***r = 0.26*** ***P < .001***	***r = 0.345*** ***P < .001***	*r = 0.124* *P = .099*					*99.95 ± 24.81/89.5 ± 27.12* ***P = .006***
LSAS Fear	*r = –0.025* *P = .737*	***r = 0.277*** ***P < .001***	***r = 0.17*** ***P = .023***	***r = 0.382*** ***P < .001***	***r = 0.384*** ***P < .001***	***r = 0.276*** ***P < .001***	***r = 0.238*** ***P = .001***	***r = 0.327*** ***P < .001***	*r = 0.116* *P = .196*	***r = 0.948*** ***P < .001***				*52.4 ± 12.92/45.17 ± 13.67* ***P = .001***
LSAS Avoidance	*r = –0.117* *P = .12*	***r = 0.274*** ***P < .001***	***r = 0.166*** ***P = .026***	***r = 0.366*** ***P < .001***	***r = 0.378*** ***P < .001***	***r = 0.258*** ***P < .001***	***r = 0.24*** ***P = .001***	***r = 0.32*** ***P < .001***	*r = 0.097* *P = .124*	***r = 0.96*** ***P < .001***	***r = 0.841*** ***P < .001***			*47.55 ± 12.62/44.33 ± 15.79* ***P = .022***
RSES	*r = –0.094* *P = .212*	*r = 0.106* *P = .157*	*r = 0.065* *P = .384*	***r = 0.221*** ***P = .03***	***r = 0.247*** ***P = .001***	*r = 0.109* *P = .147*	***r = 0.383*** ***P < .001***	***r = 0.355*** ***P < .001***	***r = 0.322*** ***P < .001***	***r = 0.419*** ***P < .001***	***r = 0.351*** ***P < .001***	***r = 0.441*** ***P < .001***		*13.51 ± 7.15/11.35 ± 7.7* *P = .058*

Bold and italic values indicate statistical significance ($P < .05$).

ASRS = Adult ADHD Self-Report Scale, LSAS = Liebowitz Social Anxiety Scale: For Adults, RSES = Rosenberg Self-Esteem Scale-Short Form, SMAS-AF = Social Media Addiction Scale-Adult Forum, VC = virtual communication, VT = virtual tolerance.

*Mann–Whitney *U* test.

Table 4

Comparison of clinical variables in groups separated by social media usage patterns.

Comparison of communication on social media with face-to-face communication	More difficult (n = 75, 41.9%)	No difference (n = 15, 19%) *n (%)*	Easier (n = 70, 39.1%)	*P**
Gender				
Female	51 (39.2%)	25 (19.2%)	54 (41.5%)	*.463*
Male	24 (49%)	9 (18.4%)	16 (32.7%)	
		Mean ± SD		*P***
Age	20.87 ± 1.77	20.29 ± 1.71	20.36 ± 1.73	*.144*
Screen time	327.65 ± 143.46	349.85 ± 163.35	381.59 ± 161.73	*.103*
Anxiety point unable to access (0–10)	4.89 ± 2.44	6.06 ± 2.5	5.93 ± 2.84	***.024***
SMAS-AF Total	53.6 ± 10.95	56.09 ± 10.47	56.57 ± 13.66	*.52*
SMAS-AF Virtual Tolerance	33.19 ± 6.85	35.09 ± 6.22	34.91 ± 7.55	*.243*
SMAS-AF Virtual Communication	20.41 ± 5.15	21 ± 5.96	21.66 ± 7.48	*.796*
ASRS Total	31.75 ± 10.77	37.09 ± 11.6	35.74 ± 12.53	***.035***
ASRS-A	16.17 ± 6.39	18.09 ± 5.32	18.53 ± 6.99	***.024***
ASRS-B	15.57 ± 5.62	19 ± 7.37	17.21 ± 6.96	*.082*
LSAS total	94.53 ± 23.78	87.32 ± 20.31	104.8 ± 28.32	***.004***
LSAS fear	49.73 ± 12.2	45.41 ± 11.48	53.71 ± 14.93	***.014***
LSAS avoidance	44.8 ± 12.43	41.91 ± 9.68	51.09 ± 15.2	***.004***
RSES	11.11 ± 7.14	13.12 ± 6.7	14.76 ± 7.49	*.103*

Using social media by hiding the identity/using a fake account	None (n = 125, 69.8%)	Present (n = 54, 30.2%) *n (%)*	*P**
Gender			
Female	85 (65.4%)	45 (34.6%)	***.035***
Male	40 (81.6%)	9 (18.4%)	
		Mean ± SD	*P****
Age	20.58 ± 1.78	20.5 ± 1.7	*.878*
Screen time	333.4 ± 143.73	398.24 ± 173.41	***.039***
Anxiety point unable to access (0–10)	5.34 ± 2.7	5.93 ± 2.52	*.168*
SMAS-AF total	53.19 ± 11.43	59.96 ± 12.13	***.000***
SMAS-AF virtual tolerance	33.16 ± 6.89	36.69 ± 6.81	***.002***
SMAS-AF virtual communication	20.03 ± 5.73	23.28 ± 6.97	***.003***
ASRS Total	33.39 ± 11.23	36.48 ± 12.86	*.103*
ASRS-A	16.84 ± 6.3	18.89 ± 6.81	***.027***
ASRS-B	16.55 ± 6.31	17.59 ± 7.25	*.359*
LSAS total	94.76 ± 26.63	102.59 ± 23.09	***.017***
LSAS fear	49.43 ± 13.97	52.8 ± 12.07	***.042***
LSAS avoidance	45.33 ± 13.46	49.8 ± 13.44	***.017***
RSES	12.45 ± 7.16	14 ± 7.72	*.163*

Bold and italic values indicate statistical significance ($P < .05$).

ASRS = Adult ADHD Self-Report Scale, LSAS = Liebowitz Social Anxiety Scale: For Adults, RSES = Rosenberg Self-Esteem Scale-Short Form, SD = standard deviation, SMAS-AF = Social Media Addiction Scale-Adult Forum, VC = virtual communication, VT = virtual tolerance.

*Pearson chi-square test.

**Kruskal–Wallis Test.

***Mann–Whitney *U* test.

were no significant differences considering, age, the anxiety level score when not able to access social media, total score of ASRS, ASRS- B scores, total score of RSES between the groups (Table 4).

4. Discussion

While previous studies have often examined ADHD, social anxiety, or self-esteem in isolation, the present study contributes to the literature by simultaneously examining these psychological domains within a single framework. This integrative approach allows for a more nuanced understanding of how multiple vulnerabilities may coexist and interact in relation to problematic social media use. An additional contribution of this study lies in the examination of specific social media usage patterns, such as preferences for online versus face-to-face communication and the use of hidden identities or fake accounts. These behavioral indicators are rarely examined alongside standardized psychological measures and may serve as clinically relevant markers of problematic social media use risk.

The mean daily screen time of 352.29 ± 156.41 minutes and the mean SMAS-AF score of 55.40 ± 11.97 indicate a high level of social media engagement in this population. This usage exceeds the reported global average of 2 hours and 20 minutes per day,[2] may indicate that PSMU is more common among university students, particularly those aged 18 to 25, than the general population. This could be attributed to the voluntary nature of the study, potentially attracting students with higher social media usage. Whereas previous studies predominantly focused on platforms such as Facebook and Twitter, our study yielded a different result.[42] Instagram, an application emphasizing visual content, was the most frequently used platform. The evolving content and features of social media platforms suggest that studies on PSMU need to be continually updated to reflect these changes.

The correlation between total scores on the Social Media Addiction Scale and ADHD symptoms supports prior research

highlighting a link between impulsivity, attentional difficulties, and problematic digital behaviors. Specifically, SMAS-AF scores were positively associated with both the inattentive and hyperactive/impulsive subscales of ASRS, suggesting that both symptom domains of ADHD may contribute to compulsive social media engagement. These results align with prior studies indicating that individuals with ADHD traits may be more susceptible to PSMU due to difficulties with self-regulation, higher levels of distractibility, and a greater tendency to seek instant gratification.[31,34,35] In a large survey, participants reported an association between addictive social media use and hyperactivity.[40] These findings align with previous research demonstrating similar associations.[11,27,31,32] In our study, screen time was positively correlated with the ASRS-A score (attention deficit symptoms); however, it was not associated with hyperactivity/impulsivity (ASRS-B). This suggests that while hyperactivity may be a risk factor for PSMU, it is not directly related to increased screen time.

The participants were evaluated according to their comparison between communication on social media and face-to-face communication and were divided into 3 groups. There was a statistically significant difference in terms of the total score of ASRS and ASRS-A, and the group that found it more difficult to communicate on social media had lower scores. Although the literature consistently demonstrates a positive association between PSMU and ADHD, the findings of the present study offer a noteworthy observation: individuals exhibiting ADHD symptoms – particularly those characterized by inattention – appear to perceive face-to-face communication as more challenging compared to communication via social media platforms. However, the limited sample sizes across the groups may have reduced the statistical power and generalizability of these findings. If participants had been categorized into binary groups (e.g., yes vs no) rather than 3, different results might have emerged. It was found that adolescents with ADHD tend to use social media more intensely, hold more accounts, and also more fake accounts.[37-39] In our study, a statistically significant difference was observed in only ASRS-A scores between the groups divided due to hiding identity/using a fake account or not. However, no significant differences were observed in total ASRS and ASRS-B scores.

Studies focusing on PSMU and ADHD often lack clinical ADHD samples.[33] As in our study, there are existing studies that screen for ADHD symptoms; however, there is a growing need for future research that compares individuals with a clinical diagnosis of ADHD to matched control groups. Consistent with the inherent limitations of cross-sectional designs, our study cannot determine the directionality of the observed associations. Specifically, it remains uncertain whether problematic social media use contributes to the exacerbation of ADHD symptoms, or if individuals exhibiting ADHD-related traits are more susceptible to developing problematic usage patterns.

In line with the existing literature, this study also found robust positive associations between social anxiety and PSMU.[43,44] Participants with higher LSAS scores, reflecting elevated levels of both social fear and avoidance, were more likely to report higher SMAS-AF scores and greater screen time. This is congruent with the hypothesis that individuals with social anxiety may overutilize social media as a compensatory strategy to navigate social interactions that they find challenging in offline contexts.[44,45,47] The virtual environment offers a sense of control and reduced risk of negative evaluation, potentially fostering a maladaptive reliance on online communication. Differences in PSMU severity across groups, categorized by their comfort levels in online versus face-to-face communication, further support this situation. Participants who found online communication easier than face-to-face interaction demonstrated significantly higher levels of social anxiety, underscoring the possibility that psychological vulnerabilities may influence communication preferences and exacerbate problematic engagement with social platforms. At the same time, those using social media by hiding their identity or using a fake account, as expected, had higher social anxiety scores.

While the current findings reveal a significant association between PSMU, screen time, and social anxiety, they do not adequately elucidate the directionality or underlying mechanisms of this relationship. It is conceivable that individuals with heightened social anxiety may turn to social media as a compensatory strategy due to impaired face-to-face communication, thereby increasing their susceptibility to problematic use. Conversely, sustained engagement with social media and prolonged screen exposure may further entrench social anxiety symptoms by reinforcing avoidance behaviors and reducing opportunities for real-world social interaction.[49-51]

The cross-sectional design of the study limits the ability to determine the directionality of the observed associations. It is plausible that these relationships are bidirectional, whereby psychological vulnerabilities such as ADHD symptoms and social anxiety increase susceptibility to problematic social media use, while excessive and maladaptive use may further exacerbate these symptoms over time.

Interestingly, the relationship between self-esteem and PSMU was less pronounced. While there was a modest correlation between SMAS-AF and RSES scores, no significant association was observed between screen time and self-esteem, and virtual communication subscale scores did not significantly correlate with self-esteem. These findings suggest that although self-esteem may play a role in the development or maintenance of PSMU, it may not be as central as ADHD symptoms or social anxiety. This is partially consistent with the dual-pathway hypothesis in the literature, which posits that low self-esteem may either predispose individuals to excessive online engagement or be negatively impacted by it due to mechanisms such as upward social comparison or identity fragmentation.[58,60]

Gender differences were observed in both PSMU and social anxiety, with female participants exhibiting significantly higher scores on SMAS-AF and LSAS. This finding aligns with several studies that report higher social media use and social anxiety symptoms in females,[53,54] though the directionality of these associations remains a matter for further investigation. Females may engage in more emotionally driven use of social media, including seeking social support and validation, which may place them at greater risk for experiencing negative psychological outcomes when these needs are unmet or inconsistently fulfilled online. Dam et al reported that women generally use social media for more than 2 hours per day and more frequently than men, which contrasts with the findings of our study.[6] In our study, there was no difference between women and men in terms of screen time. This suggests that while men and women may spend similar amounts of time on social media, women may be more susceptible to developing social media addiction.

The presence of hidden identity or fake account usage was associated with higher SMAS-AF, LSAS, and ASRS-A scores, as well as longer screen time. This behavioral pattern may reflect a tendency to experiment with alternative self-presentations due to underlying social insecurity, impulsivity, or attempts to avoid judgment in online interactions.[46-48] Such behaviors are increasingly common and may serve as an important clinical marker for PSMU risk, particularly in populations with ADHD and social anxiety traits.

The findings should also be interpreted within the sociocultural context of Turkey. In a society where interpersonal relationships and social connectedness play a central role, social media may serve as an important extension of offline social interactions. High levels of social media penetration and widespread smartphone use may further amplify both adaptive and maladaptive patterns of engagement. These cultural factors may influence the observed associations and should be considered when generalizing the findings to other contexts.

This study has limitations, including its cross-sectional design and reliance on self-report measures, which may be subject to recall bias and variations in usage patterns. The reliance on self-reported ADHD symptoms rather than clinically confirmed diagnoses may limit the clinical interpretability of the findings, as ASRS scores reflect symptom severity rather than diagnostic status. Although participants were not recruited via social media platforms, the voluntary nature of participation and the use of convenience sampling may still have introduced selection bias. Individuals with greater interest in or engagement with social media may have been more likely to participate. Furthermore, the large number of statistical tests conducted increases the risk of type I error, and the findings should therefore be interpreted with caution. Future research should employ longitudinal designs and consider more objective measures of social media usage, although social media addiction differs from other internet and internet-related addictions due to its unique characteristics.

5. Conclusion

This study has several strengths, including the use of well-validated instruments, a relatively large and gender-diverse university student sample, and the simultaneous examination of multiple psychological domains. The findings highlight the importance of psychological traits – particularly ADHD symptoms and social anxiety – in understanding problematic patterns of social media use among university students. While self-esteem was also associated with problematic social media use, its role appeared weaker and more nuanced, suggesting that its influence may be moderated by other psychological or behavioral factors.

Overall, the results support the conceptualization of problematic social media use as a multidimensional phenomenon shaped by individual vulnerabilities and usage-related behaviors rather than by time spent online alone. By providing data from a non-Western university student population, this study extends existing research beyond predominantly Western and East Asian contexts and contributes to the cultural generalizability of findings in this field. Given the pervasive role of social media across multiple domains of modern life, further interdisciplinary and longitudinal research is needed to clarify underlying mechanisms and to inform prevention and intervention strategies targeting vulnerable populations.

From a clinical perspective, these findings suggest that problematic social media use may be particularly relevant among university students presenting with ADHD symptoms and social anxiety. Incorporating screening for problematic social media use into routine mental health assessments in university settings may be beneficial. In addition, interventions targeting self-regulation, social skills, and healthy digital habits may help reduce the risk of problematic use and its associated psychological difficulties.

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